

154. B takes the value
 (a) 0 (b) 5
 (c) 9 (d) 0 or 9
155. C takes the value
 (a) 0 (b) 2
 (c) 2 or 3 (d) 5
156. A 3-digit number $4a3$ is added to another 3-digit number 984 to give the four-digit number $13b7$, which is divisible by 11. Then, $(a + b)$ is (M.B.A., 2006)
 (a) 10 (b) 11
 (c) 12 (d) 15
157. If $ab \overline{)252} \overline{)ba}$, the values of a and b are (I.A.M., 2007)
- $$\begin{array}{r} 24 \\ 12 \\ \underline{12} \\ \times \end{array}$$
- (a) 1, 2 (b) 2, 3
 (c) 1, 3 (d) None of these
- 158.
- $$\begin{array}{r} * * * \\ \times * \\ \hline 8 * * 1 \end{array}$$
- In the above multiplication problem, $*$ is equal to
 (a) 1 (b) 3
 (c) 7 (d) 9
159. If $*$ means adding 6 times the second number to the first number, then $(1 * 2) * 3$ equals
 (a) 21 (b) 31
 (c) 91 (d) 93
160. If $1 \times 2 \times 3 \times \dots \times n$ is denoted by $\lfloor n \rfloor$, then $\lfloor 8 \rfloor - \lfloor 7 \rfloor - \lfloor 6 \rfloor$ is equal to
 (a) $6 \times 7 \times \lfloor 8 \rfloor$ (b) $7 \times 8 \times \lfloor 7 \rfloor$
 (c) $6 \times 8 \times \lfloor 6 \rfloor$ (d) $7 \times 8 \times \lfloor 6 \rfloor$
161. The highest power of 9 dividing $99!$ completely is
 (a) 11 (b) 20
 (c) 22 (d) 24
162. For an integer n , $n! = n(n-1)(n-2)\dots 3.2.1$. (P.C.S., 2008)
 Then, $1! + 2! + 3! + \dots + 100!$ when divided by 5 leaves remainder
 (a) 0 (b) 1
 (c) 2 (d) 3
163. The number of prime factors in the expression $6^{10} \cdot 7^{17} \cdot 11^{27}$ is equal to
 (a) 54 (b) 64
 (c) 71 (d) 81
164. What is the number of prime factors contained in the product $30^7 \times 22^5 \times 34^{11}$?
 (a) 49 (b) 51
 (c) 52 (d) 53
165. What number multiplied by 48 will give the same product as 173 multiplied by 240?

- (a) 495 (b) 545
 (c) 685 (d) 865
166. A positive number, which when added to 1000, gives a sum which is greater than when it is multiplied by 1000. This positive integer is
 (a) 1 (b) 3
 (c) 5 (d) 7
167. 7 is added to a certain number; the sum is multiplied by 5; the product is divided by 9 and 3 is subtracted from the quotient. Thus, if the remainder left is 12, what was the original number? (S.S.C., 2005)
 (a) 20 (b) 30
 (c) 40 (d) 60
168. Symbiosis runs a Corporate Training Programme. At the end of running the first programme, its total takings were ₹ 38950. There were more than 45 but less than 100 participants. What was the participant fee for the programme? (SNAP, 2005)
 (a) ₹ 410 (b) ₹ 450
 (c) ₹ 500 (d) ₹ 510
169. The sum of four consecutive even numbers A, B, C and D is 180. What is the sum of the set of next four consecutive even numbers? (Bank Recruitment, 2008)
 (a) 196 (b) 204
 (c) 212 (d) 214
 (e) None of these
170. A young girl counted in the following way on the fingers of her left hand. She started calling the thumb 1, the index finger 2, middle finger 3, ring finger 4, little finger 5, then reversed direction, calling the ring finger 6, middle finger 7, index finger 8, thumb 9 and then back to the index figure for 10, middle finger for 11, and so on. She counted upto 1994. She ended on her
 (a) thumb (b) index finger
 (c) middle finger (d) ring finger
171. Given $n = 1 + x$ and x is the product of four consecutive integers. Then which of the following is true?
 I. n is an odd integer. II. n is prime.
 III. n is a perfect square.
 (a) Only I is correct
 (b) Only III is correct
 (c) Both I and II are correct
 (d) Both I and III are correct
172. If $x + y = 15$ and $xy = 56$, then what is the value of $x^2 + y^2$? (L.I.C.A.D.O., 2007)
 (a) 110 (b) 113
 (c) 121 (d) Cannot be determined
 (e) None of these
173. Given that $(1^2 + 2^2 + 3^2 + \dots + 20^2) = 2870$, the value of $(2^2 + 4^2 + 6^2 + \dots + 40^2)$ is
 (a) 2870 (b) 5740
 (c) 11480 (d) 28700